Beyond the Yellow Tape

Can countries with similar homicide rates have different homicide profiles?

DS608 Data Analytics for Economics, MDSE
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RESEARCH QUESTION

What we set out to answer

Can countries with similar intentional homicide rates exhibit significantly different homicide profiles?

Why it matters

Overall rates show how much lethal violence there is, but not who is affected or in what circumstances.

What we do

Map global patterns, compare victim characteristics and incident circumstances across countries, and track change over time.

What we do not claim

This is descriptive exploration. We make no causal claims about why profiles differ.

DATA SOURCE AND SERIES

UNODC Intentional Homicide Data

UNODC Data Portal

Intentional Homicide dataset

FILTERS APPLIED

- Victims of intentional homicide
- Age: total
- Years: 2015 to 2024
- Counts and rate per 100,000

Overall

- Homicide count and rate per 100,000 by country and year

Sex

- Female and male victims

Mechanism

- Firearms, another weapon, no weapon, unspecified

Relationship

- Intimate partner or family, other known, unknown

Situational context

- Organized crime, interpersonal, other, unknown

Citizenship

- National and foreign citizens

METHODOLOGY

From raw file to comparable data

1

Load and reshape

Keep total-age victims, 2015 to 2024. Counts and rates sit side by side.

2

Fill short gaps

Linear interpolation of internal gaps of up to 2 years. Filled values are flagged as imputed.

3

Derive denominators

Population = count ÷ rate. Used to derive missing category rates, flagged as derived.

4

Build profiles

Category shares for mechanism, relationship, context, citizenship and sex, for complete profiles only.

5

Explore in Streamlit

Map, similar-rate finder, two-country comparison, trends and a change explorer.

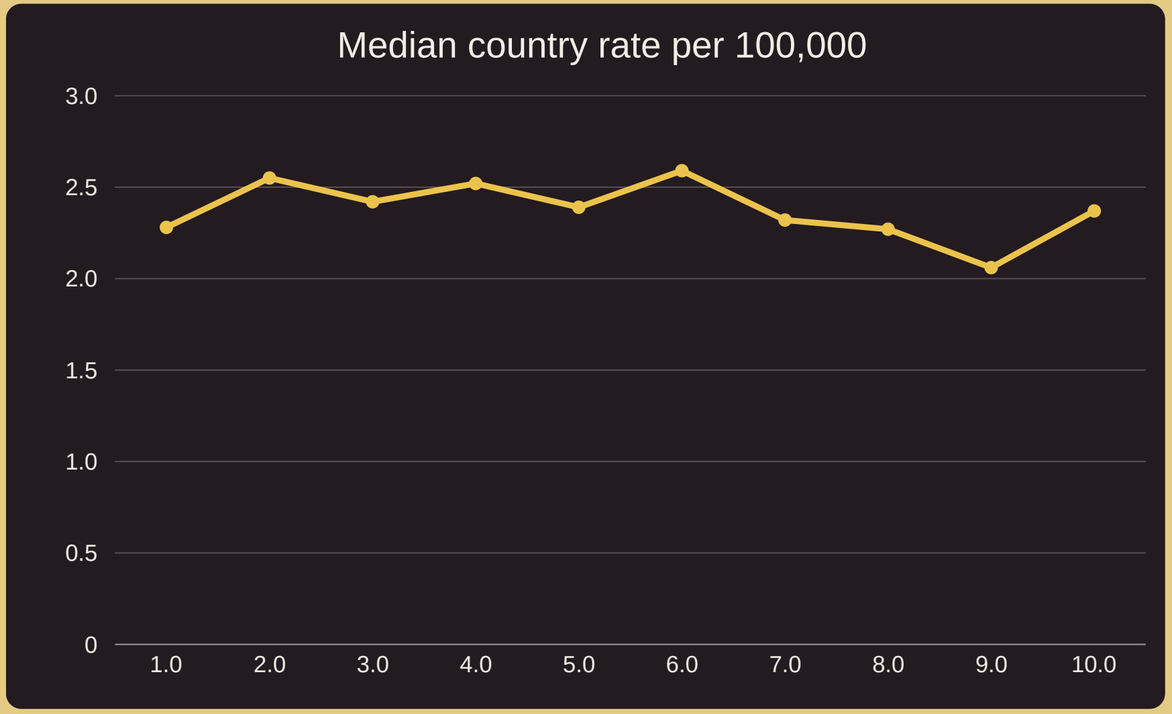
Python (pandas) wrangling → processed CSVs → Streamlit and Plotly dashboard

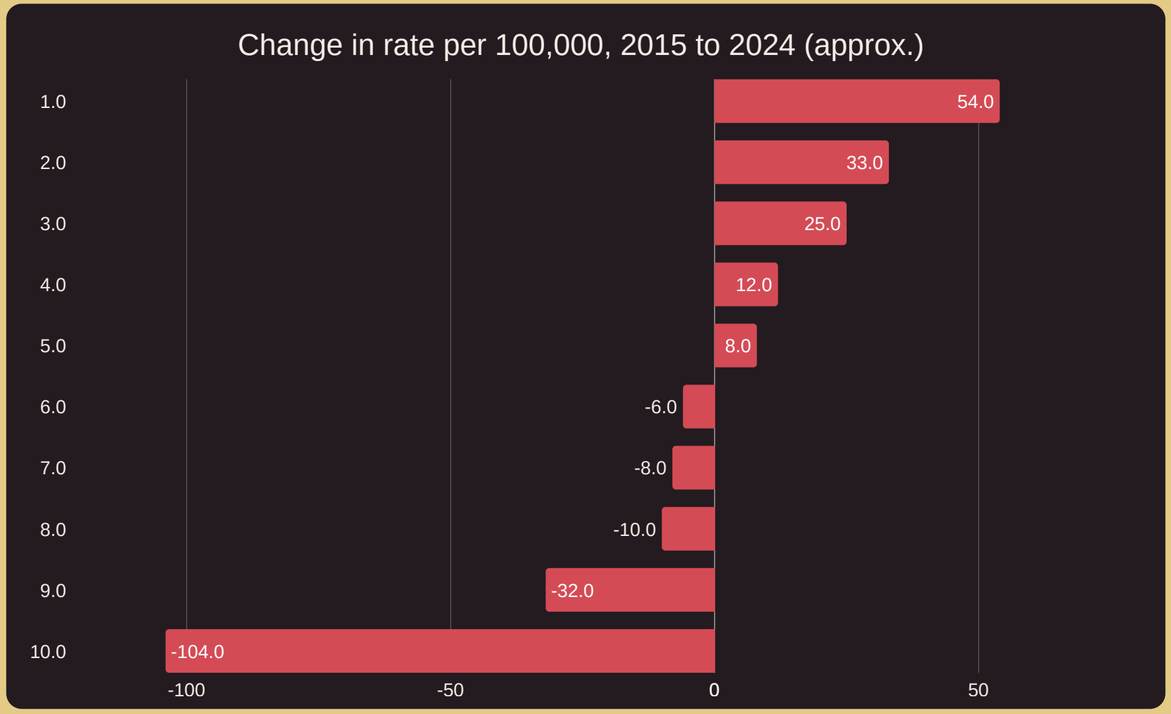
What does the Dashboard do?

1. Global Overview, map and ranking
2. Trends and “What Changed?”
3. Similar-rate country finder
4. Two-country comparison
5. Multi-dimension profile view

MAIN FINDINGS

Median rate is flat; El Salvador & Haiti extremes





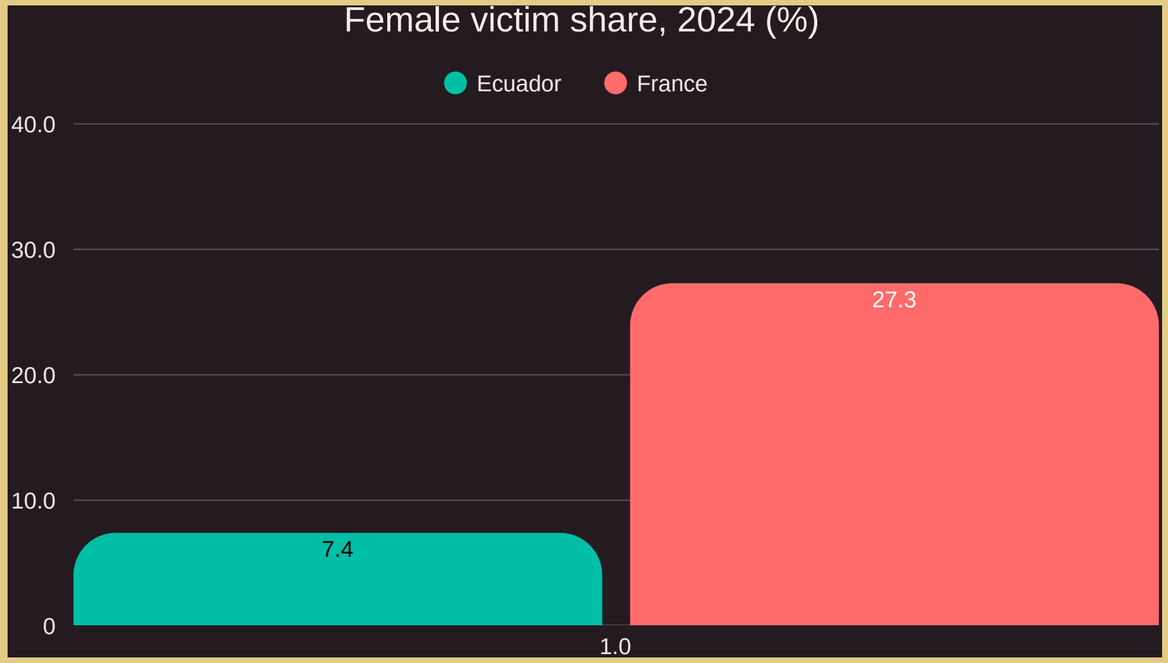
Read with care: the median stays between 2.06 and 2.59 in every year, though the number of countries with a rate falls from 144 (2019) to 95 (2024).

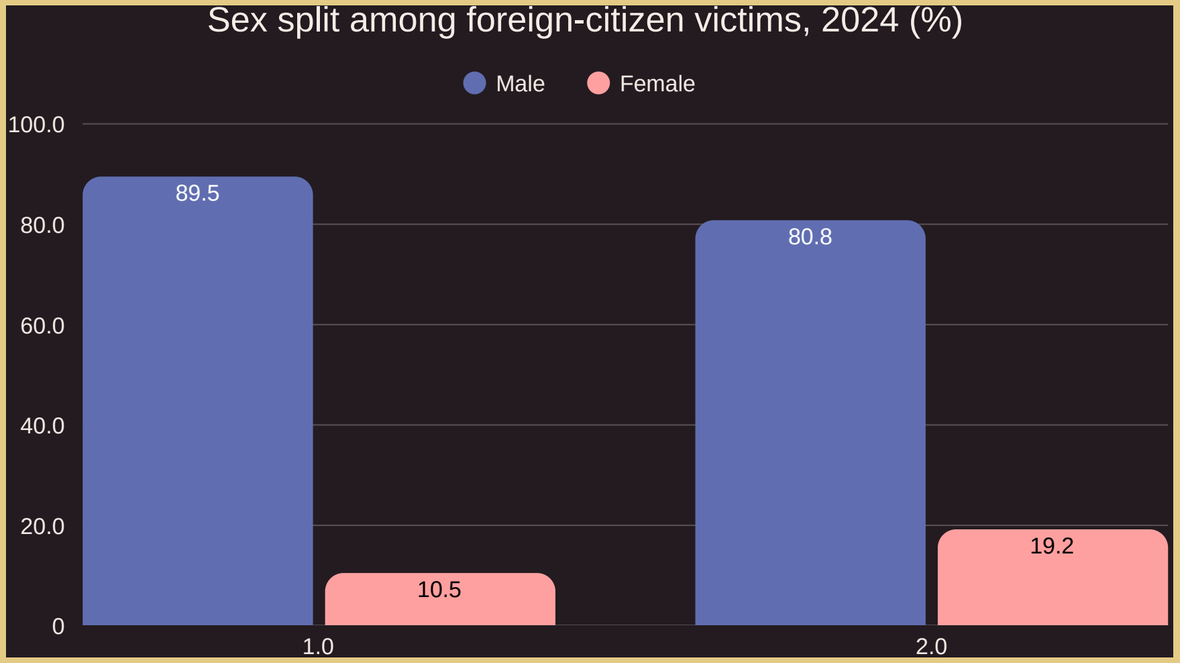
11 and 9 of the 15 largest falls and rises, respectively, are in the Americas

Absolute change in homicides per 100,000, 2015 to 2024, for countries with a rate at both ends. Top 5 shown per direction.

MAIN FINDINGS

Female share of victims is higher in France

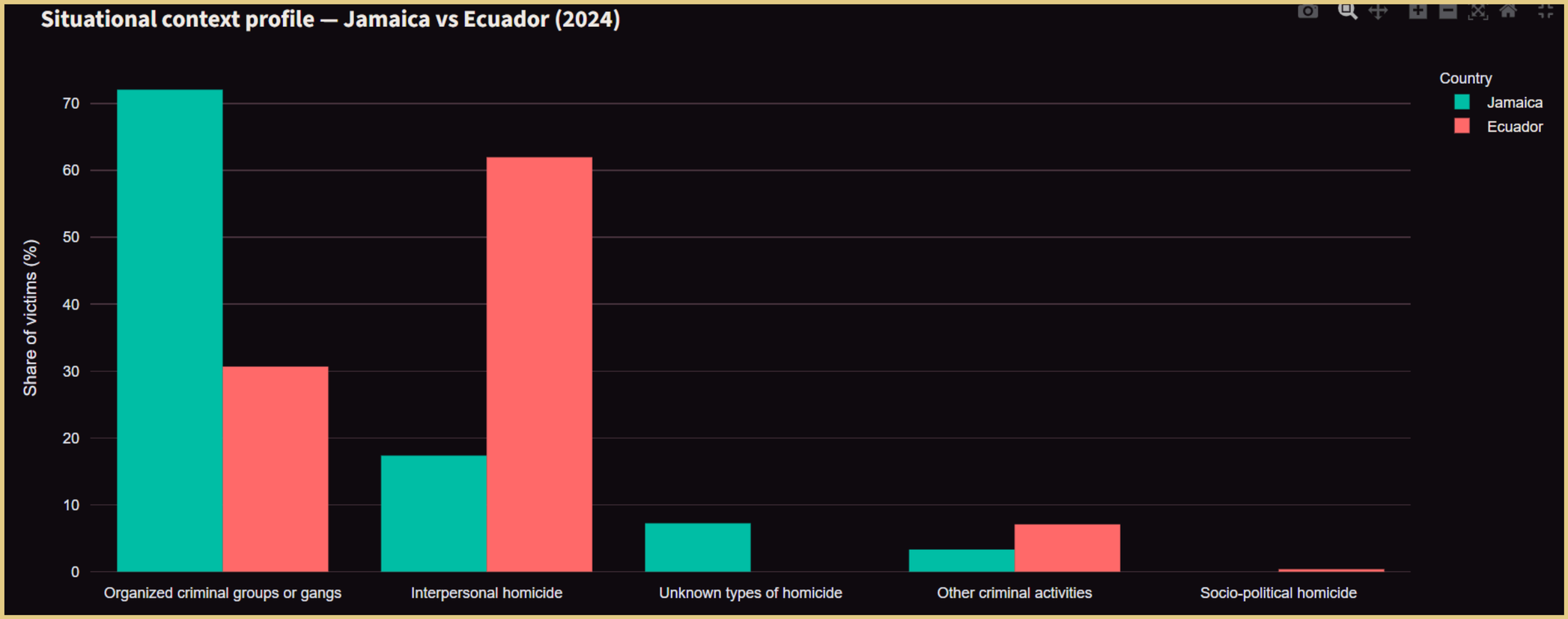




Homicide Rate (2024): Ecuador- 38.94%, France- 1.33%

Shares of victims, not female homicide rates. Fewer male victims would raise the female share even if female deaths were not higher (our inference).

Jamaica vs Ecuador profile comparison



Homicide Rate (2024): Jamaica- 40.36%, Ecuador- 38.94%

LIMITATIONS

What the results can and cannot say

Uneven coverage

Reporting standards differ, and detail is thinner than totals. Countries with a rate fall from 144 to 95.

Missing is not zero

Absent or unknown categories cannot be read as no homicides. Some profiles are almost all unknown.

Estimated values

Linear interpolation: filling gaps of up to 2 years assumes a smooth path, which can hide shocks

Shares are not risk

Real risk rate needs the female population (and the non-resident population) as the denominator. The data has only total population.

Thin dimension data

Dimension data is thinner than totals: profile comparisons rest on fewer country-years than the overall rate.

Conclusion

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Thank You!!!!

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